2025 11th International Conference on Computing and Artificial Intelligence (ICCAI) | 979-8-3315-2491-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCAI66501.2025.00088

Comparing CNN and LSTM Networks for Magnetic Localization of IoT Devices and Pedestrian Tracking

Konstantin Klipp
Daimler Center for Automotive IT Innovations (DCAITI)
Technical University Berlin
Berlin, Germany
konstantin.klipp@dcaiti.com

Edgar Blumenthal, Marten Eckardt, Jasper Windirsch
Technical University Berlin
Berlin, Germany
{e.blumenthal, m.eckardt, j.windirsch}@campus.tu-berlin.de

Bernd Schäufele, Johannes Wortmann, Ilja Radusch
Fraunhofer Institute for Open Communication Systems (FOKUS)
Berlin, Germany
{bernd.schaeufele, johannes.wortmann, ilja.radusch}@fokus.fraunhofer.de

***Abstract*—When outdoors, nowadays it is common to rely on Global Navigation Satellite Systems (GNSS) based navigation to find a location. However, for indoor environments no common solution exists, as GNSS positioning is not available indoors. While many substitute technologies rely on infrastructure installed in buildings, e.g., beacons, in this paper we use the magnetic field characteristics of buildings as a solution that is available everywhere. In the implementation, a Convolutional Neural Network (CNN) and a Long Short-Term Memory (LSTM) model are used for classification of the characteristics and regression, respectively. The approaches are evaluated against each other on a public dataset showing that the magnetic field can be a robust ubiquitous solution for indoor localization. The regression with LSTM shows the highest precision, while the error of a classification approach is constrained by the building boundaries and enables the usage of class confidence values for further processing.**

***Index Terms*—indoor positioning, neural network, IoT, magnetic field**

I. Introduction

The precise localization of persons and devices already plays a significant role in applications. It ranges from location-based services on smartphones like navigation, tracking of belongings via tags to the essential localization of robots and autonomous cars. However, GNSS signals are the main localization source, which normally require rather unobstructed line of sight to various satellites for a precise localization.

Especially in indoor environments, the GNSS accuracy is usually very low, so that no location fix can be calculated. This hinders the development and dissemination of applications. In addition to indoor navigation, which is especially important for blind users [1], IoT and robot-based applications cannot count on a reliable position indoors in public, company, or health buildings.

Hospitals, for example, benefit greatly from indoor asset tracking as important and expensive mobile equipment often gets lost in the huge buildings. They also need to track contaminated beds and equipment. Mobile emergency buttons and air quality sensors could be used in a much more flexible way if they know their own indoor position.

However, there are solutions and products to address this, but their distribution is very limited. A main reason for this issue is infrastructure costs. Many solutions rely on radio-based beacons like Bluetooth Low Energy (BLE) [2], [3], WiFi [4] or recently Ultrawideband (UWB) [5]. However, due to the high number of required beacons, installation and maintenance costs, for example, to change batteries, are often a relevant obstacle [6].

Other systems use natural or artificial optical features in combination with cameras, significantly reducing infrastructure costs [7], [8]. This comes with the drawback that visual contact and also a camera are needed, which is not applicable to smartphones in the pocket or most IoT and tracking devices.

A very promising approach is localization using Earth's magnetic field. Depending on the structure of the building, there are large metal parts in the walls and ceilings of the floors. These structures disturb the natural Earth's magnetic field in combination with other metal structures in the building. Its distribution inside the building forms a specific pattern.

Even low-cost and energy-saving Micro-Electro-Mechanical Systems (MEMS) magnetometer sensors can measure the deviations. They are built into most smartphones and are easily applicable in IoT devices. Compared to the radio-based solutions, besides the missing infrastructure costs, another advantage is that the magnetic field is much more static over time. On the other hand, it has a maximum of three features, as it is only a three-dimensional vector, and the values are not unique. Independent of the localization method, remote tracking of devices normally requires a server infrastructure, which also allows the usage of complex tracking algorithms even for IoT devices.

The same value patterns typically appear at different locations within buildings and floors. Thus, patterns can only be detected over time, respectively, the traveled distance. This makes magnetic localization a challenging task, and the algorithms are often only applicable by fusion with other localization techniques. In this paper, we present a novel standalone solution based on CNN and LSTM deep neural

979-8-3315-2491-3/25/$31.00 ©2025 IEEE
DOI 10.1109/ICCAI66501.2025.00088

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY DELHI. Downloaded on March 15,2026 at 10:45:41 UTC from IEEE Xplore. Restrictions apply.

network structures adapted to each building. We evaluate the approaches against each other in different buildings.

The remainder of this paper is structured as follows. In section II, current research in the field of indoor localization is discussed. Section III continues with the preparation of the dataset used, before the method is described in section IV. In section V, our evaluation metrics are introduced, which are used for the presentation of results in section VI. Finally, a summary and an outlook on future work are given in section VII.

II. Related Work

A. Magnetic localization

The majority of the solutions for magnetic indoor localization rely on fingerprinting techniques building up a grid-based map of the whole magnetic vector or extracted features in offline mode first. In localization mode, the classification problem is often solved with traditional algorithms, such as k-nearest neighbors [9], support vector machines (SVMs) [10], [11], and decision trees [12] for the task of identifying a localization by fingerprints [13]. They are often used for WiFi fingerprints but are mainly applicable to magnetic field data as well.

Another possibility is to use probabilistic filters, such as Bayes filters. In [14], the magnetic field is fused with a step detector comprising a gyroscope and an accelerometer in a grid-based Bayes filter showing good results. However, the solution focuses on the one-dimensional case and requires a start position which is not practical for all scenarios. Also, setting up a comprehensive magnetic field fingerprinting map can be quite challenging.

While traditional methods are less computationally expensive, deep learning approaches have recently gained significant traction and shown promising results [13], [15], [16]. CNN and Recurrent Neural Network (RNN), especially LSTM architectures, are widely used in the reviewed literature, sometimes even in combination. These architectures are particularly effective at capturing the spatial and temporal characteristics of sequential data, making them well-suited for this application. For instance, [17], [18], [19], and [20] all employed variations of RNN respectively LSTM models in their work. The complexity of these models ranges from basic RNNs [17] to more sophisticated architectures like the hierarchical LSTM proposed by [20]. [15], [16], and [13] utilized CNN-based architectures in their work.

Most of the reviewed papers directly use magnetic field patterns as input, while [16] and [21] incorporate the use of Recurrence Plots (RPs) as sequence fingerprints. In [13] the use of Temporal Convolutional Networks (TCNs) is introduced, which are specifically designed to handle sequential data by capturing long-range dependencies through a series of convolutions.

Recent research has also explored the combination of different neural network architectures to leverage their respective strengths. Both [22] and [23] use more advanced hybrid approaches that integrate CNNs with RNNs or LSTMs to build and process multichannel visual representations of magnetic data. These approaches combine the spatial processing capabilities of CNNs with the sequential learning of RNNs.

B. Dataset Selection

As the practical gathering of enough high-quality data to train the models is an arduous task, we focus on public available datasets. This also allows a comparison of our approaches with state-of-the-art algorithms. Due to the significant differences between datasets including the methods of data acquisition, the sensor frequency, and the total amount of data collected, we opted to focus on a single dataset.

All candidates offer at least 3 axis magnetometer data. The approach in [24] combines magnetic and WIFI data in 5 buildings with a recording frequency of $10\,\mathrm{Hz}$. In [25] Bluetooth data is added, covering about $2000\,\mathrm{m}^2$ on parts of three floors in a building. Additionally, the system in [26] covers around $260\,\mathrm{m}^2$ of space at $10\,\mathrm{Hz}$ as well.

For our training and evaluation, we choose the MagPie dataset as provided in [27]. It includes magnetometer data from three differently sized buildings with a total of over $960\,\mathrm{m}^2$ floor space at a $50\,\mathrm{Hz}$ sampling rate. This is advantageous as it provided multiple large locations with the same level of data quality, eliminating the need for data alignment. The MagPie dataset was collected using a smartphone for measuring the magnetometer data and a Lenovo tablet using the Tango API to collect the ground truth measurements.

There are separate measurements with the devices either handheld or attached to a robot. As the devices were mounted close to the engines of the robot, we assume corresponding artifacts in the magnetic measurements in the robot data. In order to avoid errors through this kind of artifacts, we only used the handheld data. MagPie already provides over 110 trials of walked trajectories for each building. This makes comparing different approaches easier and eliminates the need to create artificial trajectories. Also, it is close to a practical application of the proposed algorithm where a visual localization system, for example, can record magnetometer data trails of many users during their usage.

III. Data preparation

Like most existing magnetic solutions, we focus on data sequences instead of single data points as the magnetic field value is not unique at each position. The sequence length remains as a parameter for evaluation, as its effect is of high relevance for the performance and for the possible applications as well. A sliding window approach using overlapping windows is implemented (Figure 1). In contrast to consecutive, non-overlapping ones, this avoids losing information between windows, as the target position always corresponds to the last data point in the sequence. In this work we use the whole raw magnetic field vector $B = [B_x, B_y, B_z]$ accepting that this does not lead to a fully rotation-invariant solution as presented in [14]. But at the same time, we do not lose information this way, as the vector direction can also include hints about the walking direction the network can learn.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY DELHI. Downloaded on March 15,2026 at 10:45:41 UTC from IEEE Xplore. Restrictions apply.

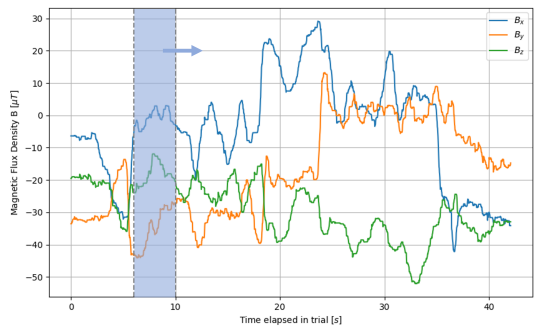

Fig. 1. Sample magnetic field data forming a specific pattern and sliding window size based on best performing sequence length for the Talbot building.

For the classification approaches, we introduce an additional step of dividing the floor plan into discrete areas to gather the corresponding landmark indices of each point. This is performed using DBSCAN clustering to segment the points and then assigning each group of points to a specific landmark/square ID of the same size. This allows us to convert the positioning problem into a classification task, where each grid cell represents a distinct location class as the target.

To structure the MagPie dataset for training, we extract the data into CSV format by matching the ground-truth positional coordinates to the magnetometer data based on the timestamps. A *trial number* column is also added to the dataset to clearly identify and separate individual trials. This allows efficient splitting of the data into training, validation, and testing sets, ensuring that no data points from the same trials appear in multiple sets. This separation is important because the data is recorded at $50\,\mathrm{Hz}$, which can lead to consecutive measurements that are very similar. If data points from the same trial appear in both training and validation sets, the model might learn to memorize these points, leading to a low validation error but poor generalization performance in real-world applications. It also helps us to later evaluate the model based on individual trials.

For classification tasks, it is also important to ensure that all classes inside the validation and test sets have corresponding targets inside the train set. Otherwise, targets that were not part of the model's training process would need to be predicted.

The inputs are of shape `(# input sequences, sequence length, 3)`. For the regression case the target is an array of shape `(# input sequences, 2)` as we predict 2-dimensional coordinates in this work. For the classification case we get the output shape `(# input sequences, 1)`. Both cases are further described in the following section.

IV. Concept and Model definition

Our method does not focus on a general model to solve the localization problem in every building. The magnetic distribution patterns can vary strongly between buildings, depending mainly on their construction method. So, we decided to keep the models as flexible as possible and train them individually for each building. We just set some very basic model structures for each approach. For each building, we identified the best model architecture and optimal training parameters by conducting a hyperparameter search using Bayesian optimization on an undersampled training set.

In our work, we focus on the usage of CNN and LSTM model architectures. In general, CNNs apply filters with different kernel sizes to the input signal detecting patterns in the signal which basically fits our problem. For the CNN variant, our model is formed by different one-dimensional convolution layers as shown in Figure 2. The number of filters, the kernel sizes as well as the number of layers are kept as a parameter for the hyperparameter tuning for each building. However, CNNs are most effective for local patterns where historical data correlations are not that relevant.

LSTMs on the other side are widely used especially for time series problem tasks. Every LSTM unit forms a memory cell to retain information over time utilizing an input-gate, an output-gate, and a forget-gate. The architecture allows the network to remember information even over longer time periods. Our LSTM integration is quite similar to the CNN version where the number of layers and the LSTM units are individually set by the Bayesian optimizer for each building. A final dense layer before the output layer is optimized as well for both model types. All models utilize the Adam optimizer. To enhance generalization within a building, dropout layers with optimizable dropout rates and L2 regularization are incorporated across all architectures. A building-specific example model for both architectures is shown in Figure 2.

We investigated two main approaches to finally predict the 2D positions. Each of the approaches are tested with a CNN and an LSTM model structure.

A. Classification approach

The concept of the classification problem approach is based on a Time Series Classification (TSC) task which is used to predict class labels for unlabeled time series of raw datapoints. The key component of TSC is to identify relevant patterns in the input data that allow for further classification which is applicable to our problem. Hence, the localization space will be divided into discrete areas with landmark indices. The classification models employ sparse categorical cross-entropy loss and a softmax activation.

B. Regression approach

In the other approach, we treat the prediction as a regression problem. The idea is based on Time Series Regression (TSR), which is used to predict continuous values based on time series data. The method is often employed for forecasting purposes. With Time Series Extrinsic Regression (TSER) as a specific type of TSR, we predict a continuous response variable that is not directly related to the time series being analyzed [28], [29]. Thereby, we directly predict the position vector $P = [P_x, P_y]$ in our case. The models use a mean squared error loss function,

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY DELHI. Downloaded on March 15,2026 at 10:45:41 UTC from IEEE Xplore. Restrictions apply.

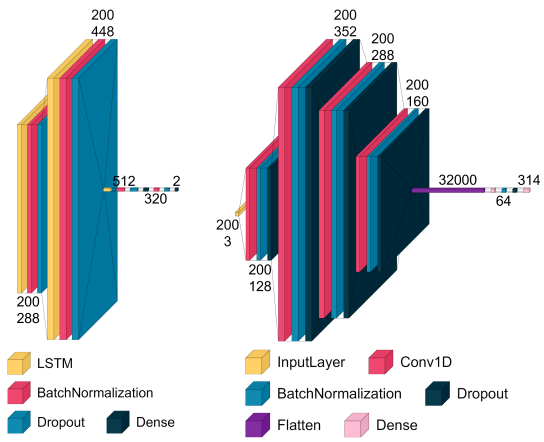

Fig. 2. Example models for the CSL building using the TSER approach with LSTM (left) and the TSR approach with CNN architecture (right) [30].

where the output layer consists of two units to directly predict the x and y coordinates.

V. Evaluation Metrics

For regression tasks, we use the Mean Absolute Error (MAE) between the ground truth and the predicted positional coordinates to assess and compare the performance of the final model. The MAE is given by the following formula:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{1}$$

where y_i are the ground truth coordinates and $\hat{y}_i$ are the predicted coordinates.

For classification tasks, the evaluation is more complex. The typical metric is accuracy, which indicates the percentage of cases in which the model correctly predicts the landmark index. However, this metric has limitations:

1) It does not really help with the comparison between models trained on different grid square sizes.
2) It does not provide information on the distances between predicted and actual points when the model misclassifies a point, making it difficult to compare with the regression case.

To address these issues, we introduced a distance-based metric using an adjacency matrix that holds the distances between landmark IDs. This metric computes the Euclidean distance between the centroids of predicted and actual landmarks, allowing us to basically mimic the mean absolute error between the centroids of landmarks. All distances are reported in meters.

VI. Results

A. Comparing LSTM and CNN in Regression Mode

We first examined the performance difference between the LSTM and CNN architectures using the same sequence lengths to have a basis for comparing both approaches. As shown in Table I for a sequence length of 50 samples, the LSTM architecture achieved an MAE of 3.12 m, whereas the CNN had an MAE of 4.41 m. When further increasing the sequence length to 100 and 200, the LSTM approach starts to significantly outperform the CNN by even reducing the MAE to 0.82 m compared to 2.31 m for a sequence length of 200. This can be seen even better when visually comparing the prediction performance, as shown in Figure 3, the LSTM model predicts the trajectory with much higher accuracy, closely following the ground truth. In contrast, the CNN model shows considerable fluctuations and deviations from the actual path, even though both models receive exactly the same information.

TABLE I
TSER performance in Talbot building. LSTMs outperform CNNs for every sequence length on test set data.

Model Type	Sequence Length	MAE
LSTM	50	3.12
CNN	50	4.41
LSTM	100	1.59
CNN	100	2.58
LSTM	200	0.82
CNN	200	2.31

In other evaluation runs, we especially focused on the sequence lengths itself. While the error goes down when increasing the context from 50 the way up to 200 samples, beyond this, the performance started to decline, with the Mean Absolute Error (MAE) increasing slightly for longer sequences (e.g., 0.9 m at 250, 0.96 m at 300, and 1.07 m at 500) in the Talbot building. These results suggest that shorter sequences provided insufficient context for accurate position prediction, leading to higher errors. Longer sequences can make the predictions more accurate, but also introduce challenges that can lead to an increase in error. This likely comes from the fact that longer sequences introduce much more possible trajectories that could lead to the same point, which requires diverse training data that includes all those possible trajectories.

B. Comparing LSTM and CNN in Classification Mode

Unlike the regression case, there is no clear better architecture when comparing CNN and LSTM approaches. Their performance is nearly equal across different configurations. Interestingly, while CNNs seem to outperform LSTM for lower sequence lengths, LSTMs are slightly superior for larger sequence lengths. This trend is visible in Table II. Additionally, the grid size plays a role in the TSC case where smaller grid sizes lead to a worse class-accuracy but also a lower MAE, which is more important in this case. The contradiction is due to the trade-off between classification accuracy and spatial precision. With larger grid sizes, the model more often predicts the correct square, increasing accuracy. However, mistakes lead to bigger spatial errors. For example, with a 2-meter grid, a wrong prediction in the neighboring cell automatically causes a 2 m error as shown in Figure 4. On the other hand, smaller grid sizes, do not achieve particularly high accuracies.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY DELHI. Downloaded on March 15,2026 at 10:45:41 UTC from IEEE Xplore. Restrictions apply.

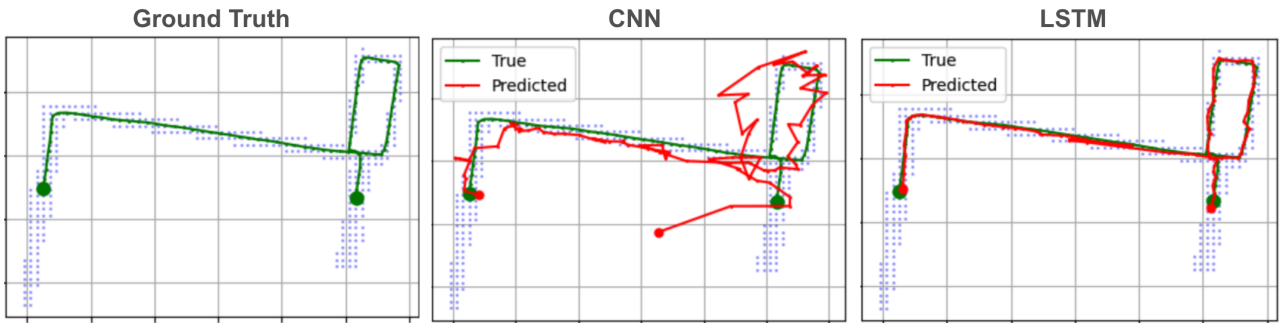

Fig. 3. Comparison of models using a sequence length of 200 in the same Talbot building trial in regression mode. LSTM models perform much better here.

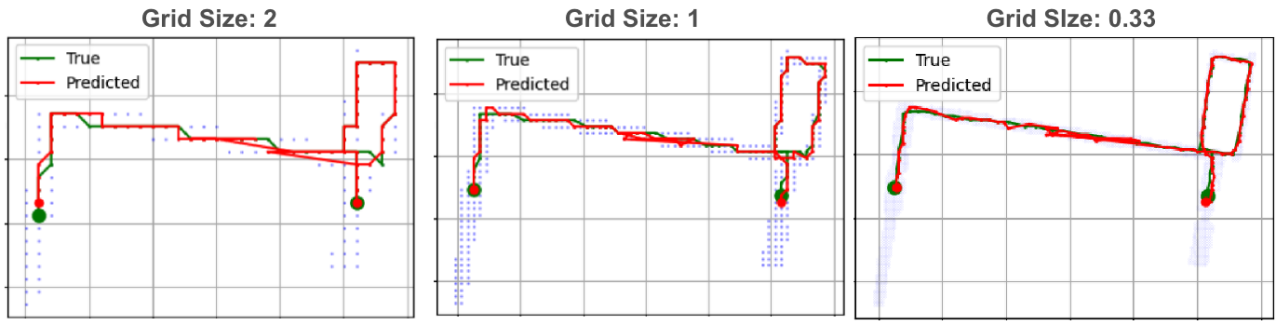

Fig. 4. Example TSC performance for different grid sizes with a sequence length of 200 on the same test set trial. Lower grid sizes lead to a lower accuracy but also reduce the MAE.

Though, when predictions are incorrect, most of the time they are spatially closer to the true position, resulting in lower overall errors.

However, it is important to note that blindly decreasing the grid size is not a viable strategy, as besides the computational increase, the amount of available training data in each grid cell decreases. So, similar to the sequence length described in subsection VI-A, the amount of data limits the minimal grid size and so effects the accuracy respectively the MAE. Regarding the given data for the Talbot building, we found an optimal grid size of 0.33 m with a sequence length of 200 samples leading to an accuracy of 0.21 and a very good MAE of 1.04 m.

TABLE II
TSC PERFORMANCE IN TALBOT BUILDING COMPARING CNN AND LSTM ARCHITECTURES. CNN AND LSTM PERFORM SIMILAR WHILE GRID SIZE AND SEQUENCE LENGTH HAVE MAJOR IMPACT.

Model Type	Sequence Length	Grid Size	Accuracy	MAE
CNN	50	2.0	0.65	7.96
LSTM	50	2.0	0.57	10.15
CNN	100	2.0	0.70	5.31
LSTM	100	2.0	0.68	5.42
CNN	300	1.0	0.52	2.37
LSTM	300	1.0	0.60	2.29
CNN	300	0.5	0.34	1.42
LSTM	300	0.5	0.39	1.34

C. Performance of TSC and TSER across buildings

Comparing the TSC and TSER, the regression approach generally outperformed the classification method in terms of positioning accuracy and precision. Our TSER method consistently achieved a lower MAE across various configurations. The regression approach offers a more straightforward implementation, as it does not require dividing the building into distinct squares. This simplicity extends to benchmarking, as there are fewer possible configurations, and the results are directly comparable via MAE. However, the classification approach has a unique advantage: Predictions are always constrained within the building's boundaries, which is not guaranteed with regression. Additionally, the TSC approach is able to output class confidences, which can be a useful information if the results will be fused with other location sources in a later processing stage.

For both approaches, the sequence length is a crucial parameter whose optimal value can vary across buildings. Most of the time, the optimal performance was achieved with sequence lengths of around 200 datapoints for the CSL and Talbot buildings, which relates to 4s with a sensor frequency at 50Hz. However, larger buildings or datasets with more extended trajectories might benefit from even longer sequences, as seen in the Loomis building results, where performance continued to improve even up to a sequence length of 500.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY DELHI. Downloaded on March 15,2026 at 10:45:41 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISON OF MAGNETOMETER LOCALIZATION APPROACHES FROM [22] AGAINST OURS.

Building	CNN+FN		CNN+RNN	Ours (MAE)	
	Landmarks	Regr.	Regr.	Landmarks	Regr.
CSL	0.95	0.97	0.30	0.24	**0.22**
Talbot	4.49	4.72	1.06	1.04	**0.82**
Loomis	4.62	5.05	**1.07**	2.48	1.98

D. Final Results and Comparison

As the magnetic localization is a not a classical machine learning research field and its performance depends on the data respectively the buildings, a comparison between solutions of different researchers is quite challenging. The differences in the rare public datasets and the fact that many state-of-the-art solutions use their own datasets makes it even more difficult. However, [22] compared their different approaches on the MagPie dataset. They also compared regression vs. classification approaches using a combination of CNN and Fully Connected Networks (FN) as well as Recurrent Neural Networks (RNN). Be aware that it stays unclear whether they also used the provided test dataset to evaluate their models or if they conducted their own test splits. Table III provides the comparison of the different approaches outlined in the paper against our approach. Note that [22] did not mention which metric they exactly used in their reported results. Thus, the comparison should therefore rather be regarded as an indication.

Comparably, our results show better overall performance for the Talbot and CSL buildings for both regression and classification, but worse overall performance for the Loomis building. It needs to be mentioned that [22] achieved the best results using RNN which require a starting point. Although they added noise of 2-3 meters to the starting point, the results are not fully comparable for the RNN version with our more challenging lost robot problem approach. On the other hand, as [22] differs in their approach and especially their RNN solution performs on the same level, we can assume that it might also perform better in other buildings. Despite this, the discrepancy for the Loomis building could also be due to their use of longer sequences (about 7 - 10 seconds) which was not in our scope. If we just compare both approaches regarding the lost robot problem having no start position even our landmark-based solution outperformed their CNN+FN approach in all three buildings. It is important to note that for our results in this section, we always report the best overall outcomes, independent of architecture, sequence length, and grid size.

VII. CONCLUSION

In this paper, we presented an approach based on Deep Learning for indoor localization, without the need of additional infrastructure. To this end, the specific magnetic field characteristics of buildings are exploited. For this purpose, the magnetic field inside a building is measured along with ground truth position data forming a set of single trails. For each building, we trained two different types of neural networks, a CNN and a LSTM, for two different localization tasks each. The first approach treats the localization task as a classification problem, while the other approach formulates the problem as a regression task.

We have then optimized the network architecture and model parameters for both network types and localization tasks on a per-building-level. The localization accuracy is thoroughly evaluated in terms of key parameters, network type, and task formulation. While treating the localization task as a classification problem does not reach the high precision of the regression task, it has the advantage of constraining predictions within building boundaries. Our optimized models reach similar or even better results than the much more complex models described in [22].

To enable a practical implementation, we plan to validate our findings on larger datasets that also include large vertical displacements (such as floor changes) and a wider range of building types and user movements. In the next step, the localization could then be integrated in a mobile indoor navigation app, such as everGuide [1] or an IoT-asset tracking use-case. If required by the use-case, the precision could be increased by spatial context between individual predictions or by leveraging class probabilities of all landmark indices, instead of relying just on the best one. Additionally, more network architectures or modules used during the hyperparameter tuning, are worth evaluating in the future.

REFERENCES

[1] J. Wortmann, B. Schaeufele, K. Klipp, I. Radusch, K. Blaß, and T. Jung, "Enhanced Accessibility for Mobile Indoor Navigation," in *2024 14th International Conference on Indoor Positioning and Indoor Navigation (IPIN)*. IEEE, 2024.

[2] F. J. Aranda, F. Parralejo, T. Aguilera, F. J. Álvarez, and J. Torres-Sospedra, "RSS Channel-Based Integration for BLE Fingerprinting Positioning," in *2023 13th International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, 2023, pp. 1–6.

[3] L. Bouse, S. A. King, and T. Chu, "Simplified Indoor Localization Using Bluetooth Beacons and Received Signal Strength Fingerprinting with Smartwatch," *Sensors*, vol. 24, no. 7, 2024. [Online]. Available: https://www.mdpi.com/1424-8220/24/7/2088

[4] J. Hu and C. Hu, "A WiFi Indoor Location Tracking Algorithm Based on Improved Weighted K Nearest Neighbors and Kalman Filter," *IEEE Access*, vol. 11, pp. 32 907–32 918, 2023.

[5] J. Qu, "A review of UWB indoor positioning," *Journal of Physics: Conference Series*, vol. 2669, no. 1, p. 012003, dec 2023. [Online]. Available: https://dx.doi.org/10.1088/1742-6596/2669/1/012003

[6] R. S. Naser, M. C. Lam, F. Qamar, and B. B. Zaidan, "Smartphone-Based Indoor Localization Systems: A Systematic Literature Review," *Electronics*, vol. 12, no. 8, 2023. [Online]. Available: https://www.mdpi.com/2079-9292/12/8/1814

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY DELHI. Downloaded on March 15,2026 at 10:45:41 UTC from IEEE Xplore. Restrictions apply.

[7] Q. Zeng, B. Ou, R. Wang, H. Yu, J. Yu, and Y. Hu, "A Robust Indoor Localization Method Based on DAT-SLAM and Template Matching Visual Odometry," *IEEE Sensors Journal*, vol. 23, no. 8, pp. 8789–8796, 2023.

[8] F. Gao and J. Ma, "Indoor Location Technology with High Accuracy Using Simple Visual Tags," *Sensors*, vol. 23, no. 3, 2023. [Online]. Available: https://www.mdpi.com/1424-8220/23/3/1597

[9] D. Li, B. Zhang, and C. Li, "A Feature-Scaling-Based k-Nearest Neighbor Algorithm for Indoor Positioning Systems," *IEEE Internet of Things Journal*, vol. 3, no. 4, pp. 590–597, 2016.

[10] A. S. Abdou, M. A. Aziem, and A. Aboshosha, "An efficient indoor localization system based on Affinity Propagation and Support Vector Regression," in *2016 Sixth International Conference on Digital Information Processing and Communications (ICDIPC)*, 2016, pp. 1–7.

[11] A. Chriki, H. Touati, and H. Snoussi, "SVM-based indoor localization in Wireless Sensor Networks," in *2017 13th International Wireless Communications and Mobile Computing Conference (IWCMC)*, 2017, pp. 1144–1149.

[12] J. Yim, "Introducing a decision tree-based indoor positioning technique," *Expert Syst. Appl.*, vol. 34, no. 2, p. 1296–1302, Feb. 2008. [Online]. Available: https://doi.org/10.1016/j.eswa.2006.12.028

[13] G. Ouyang, K. Abed-Meraim, and Z. Ouyang, "Magnetic-field-based indoor positioning using temporal convolutional networks," *Sensors*, vol. 23, no. 3, p. 1514, 2023.

[14] K. Klipp, H. Rosé, J. Willaredt, O. Sawade, and I. Radusch, "Rotation-invariant magnetic features for inertial indoor-localization," in *2018 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*. IEEE, 2018, pp. 1–10.

[15] I. Ashraf, M. Kang, S. Hur, and Y. Park, "MINLOC: Magnetic field patterns-based indoor localization using convolutional neural networks," *IEEE Access*, vol. 8, pp. 66 213–66 227, 2020.

[16] M. Abid, P. Compagnon, and G. Lefebvre, "Improved CNN-based magnetic indoor positioning system using attention mechanism," in *2021 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*. IEEE, 2021, pp. 1–8.

[17] H. J. Jang, J. M. Shin, and L. Choi, "Geomagnetic field based indoor localization using recurrent neural networks," in *GLOBECOM 2017-2017 IEEE Global Communications Conference*. IEEE, 2017, pp. 1–6.

[18] H. J. Bae and L. Choi, "Large-scale indoor positioning using geomagnetic field with deep neural networks," in *ICC 2019-2019 IEEE International Conference on Communications (ICC)*. IEEE, 2019, pp. 1–6.

[19] B. Bhattarai, R. K. Yadav, H.-S. Gang, and J.-Y. Pyun, "Geomagnetic field based indoor landmark classification using deep learning," *IEEE Access*, vol. 7, pp. 33 943–33 956, 2019.

[20] L. Wang, H. Luo, Q. Wang, W. Shao, and F. Zhao, "A hierarchical LSTM-based indoor geomagnetic localization algorithm," *IEEE Sensors Journal*, vol. 22, no. 2, pp. 1227–1237, 2021.

[21] N. Lee, S. Ahn, and D. Han, "AMID: Accurate magnetic indoor localization using deep learning," *Sensors*, vol. 18, no. 5, p. 1598, 2018.

[22] L. Antsfeld and B. Chidlovskii, "Magnetic field sensing for pedestrian and robot indoor positioning," in *2021 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*. IEEE, 2021, pp. 1–8.

[23] L. Fernandes, S. Santos, M. Barandas, D. Folgado, R. Leonardo, R. Santos, A. Carreiro, and H. Gamboa, "An infrastructure-free magnetic-based indoor positioning system with deep learning," *Sensors*, vol. 20, no. 22, p. 6664, 2020.

[24] I. Ashraf, S. Din, M. U. Ali, S. Hur, Y. B. Zikria, and Y. Park, "MagWi: Benchmark Dataset for Long Term Magnetic Field and Wi-Fi Data Involving Heterogeneous Smartphones, Multiple Orientations, Spatial Diversity and Multi-Floor Buildings," *IEEE Access*, vol. 9, pp. 77 976–77 996, 2021.

[25] Z. Tóth and J. Tamás, "Miskolc IIS hybrid IPS: Dataset for hybrid indoor positioning," in *2016 26th International Conference Radioelektronika (RADIOELEKTRONIKA)*, 2016, pp. 408–412.

[26] J. Torres-Sospedra, D. Rambla, R. Montoliu, O. Belmonte, and J. Huerta, "UJIIndoorLoc-Mag: A new database for magnetic field-based localization problems," in *2015 International conference on indoor positioning and indoor navigation (IPIN)*. IEEE, 2015, pp. 1–10.

[27] D. Hanley, A. B. Faustino, S. D. Zelman, D. A. Degenhardt, and T. Bretl, "MagPIE: A dataset for indoor positioning with magnetic anomalies," in *2017 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*. IEEE, 2017, pp. 1–8.

[28] D. Guijo-Rubio, M. Middlehurst, G. Arcencio, D. F. Silva, and A. Bagnall, "Unsupervised feature based algorithms for time series extrinsic regression," *Data Mining and Knowledge Discovery*, pp. 1–45, 2024.

[29] N. Mohammadi Foumani, L. Miller, C. W. Tan, G. I. Webb, G. Forestier, and M. Salehi, "Deep learning for time series classification and extrinsic regression: A current survey," *ACM Computing Surveys*, vol. 56, no. 9, pp. 1–45, 2024.

[30] P. Gavrikov, "visualkeras," https://github.com/paulgavrikov/visualkeras, 2020.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY DELHI. Downloaded on March 15,2026 at 10:45:41 UTC from IEEE Xplore. Restrictions apply.